# 3GPP Long Term Evolution

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## Outline

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2. Introduction II — Constellation Morph
3. 1.2 — OFDM Modulation
4. 1.5 — OFDM Explorer
5. 2.2 — LTE Bandwidth vs Resource Blocks
6. 2.3 — LTE Duplexing
7. 2.6 — OFDMA Downlink Scheduler
8. 2.7 — Different Modulation Schemes
9. 3.2 — Transmission Reception Layer

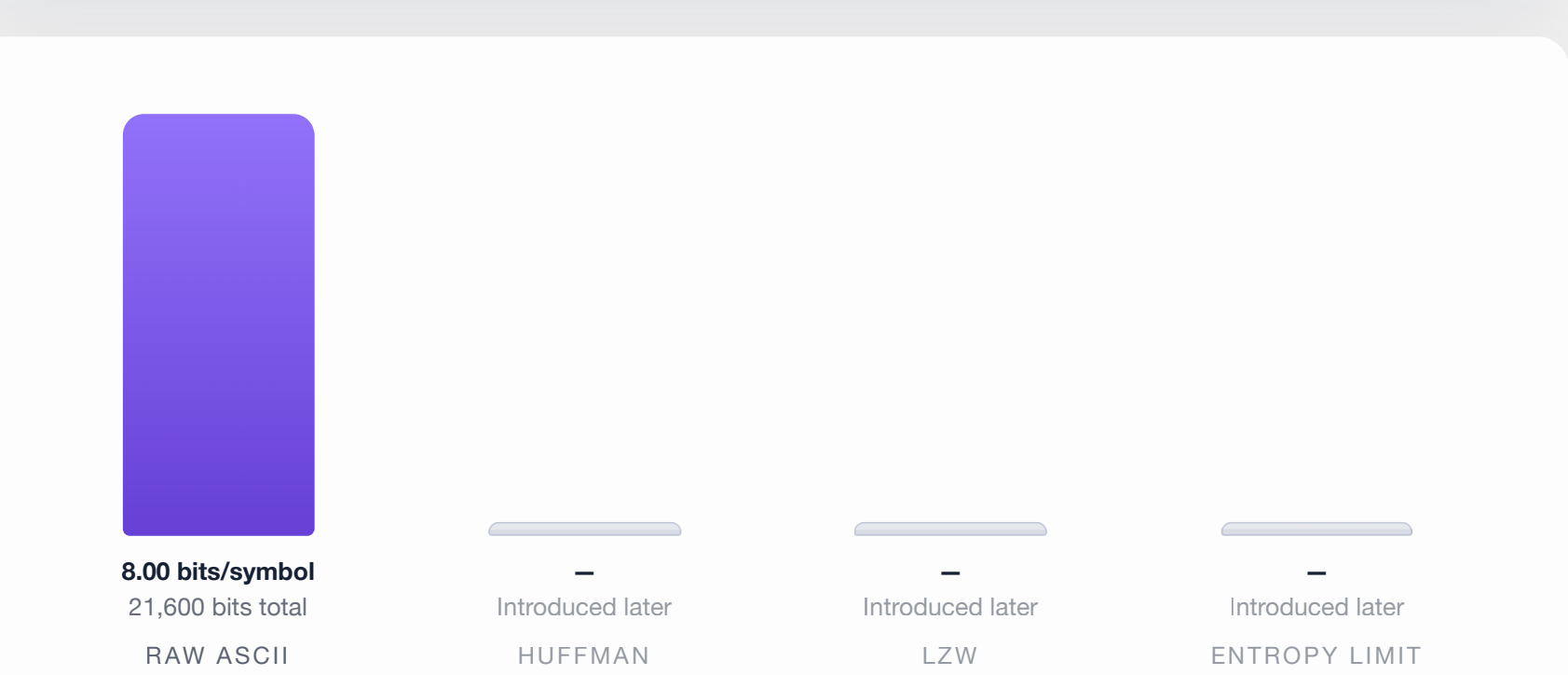
> The visualizations below are interactive in the browser.
> View the full interactive version at: **ariecr.github.io/Project_B/combined_visualizations.html**

## Introduction I

# Following Compression Down to the Entropy Bound

Watch a simple text stream shrink from raw 8-bit ASCII through Huffman coding and then Lempel–Ziv–Welch (LZW) dictionary coding. Each step pushes the average bits per symbol, illustrating how Shannon's First Theorem sets the target for a memoryless source and how richer source models move that limit.

| SYMBOLS ANALYSED | DISTINCT SYMBOLS | RAW SIZE | HUFFMAN SIZE | LZW SIZE |
|---|---|---|---|---|
| 2,700 | 15 | 21,600 bits | 9,600 bits | 4,164 bits |



1 Raw stream · 2 Huffman coding · 3 LZW dictionaries

STAGE 1 OF 3

**Raw ASCII stream**

2,700 symbols × 8 bits = 21,600 bits in fixed-width storage.

Play animation · Step › · Reset

### Experiment with your own text

Paste any sample text below. The entropies, compression efficiencies, and bar chart update as you type.

```
THE RAIN IN SPAIN FALLS MAINLY IN THE PLAIN. THE RAIN IN SPAIN FALLS MAINLY IN THE PLAIN. THE
RAIN IN SPAIN FALLS MAINLY IN THE PLAIN. THE RAIN IN SPAIN FALLS MAINLY IN THE PLAIN. THE RAIN
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MAINLY IN THE PLAIN. THE RAIN IN SPAIN FALLS MAINLY IN THE PLAIN. THE RAIN IN SPAIN FALLS
```

Update chart · Use default sample

Repetition Drill — Loops a single sentence so Huffman drops sharply and LZW nearly reaches the entropy bound by caching whole phrases. • 2,700 symbols · 15 distinct

CURATED EXAMPLES

**Repetition Drill**
Loops a single sentence so Huffman drops sharply and LZW nearly reaches the entropy bound by caching whole phrases.
Use this text

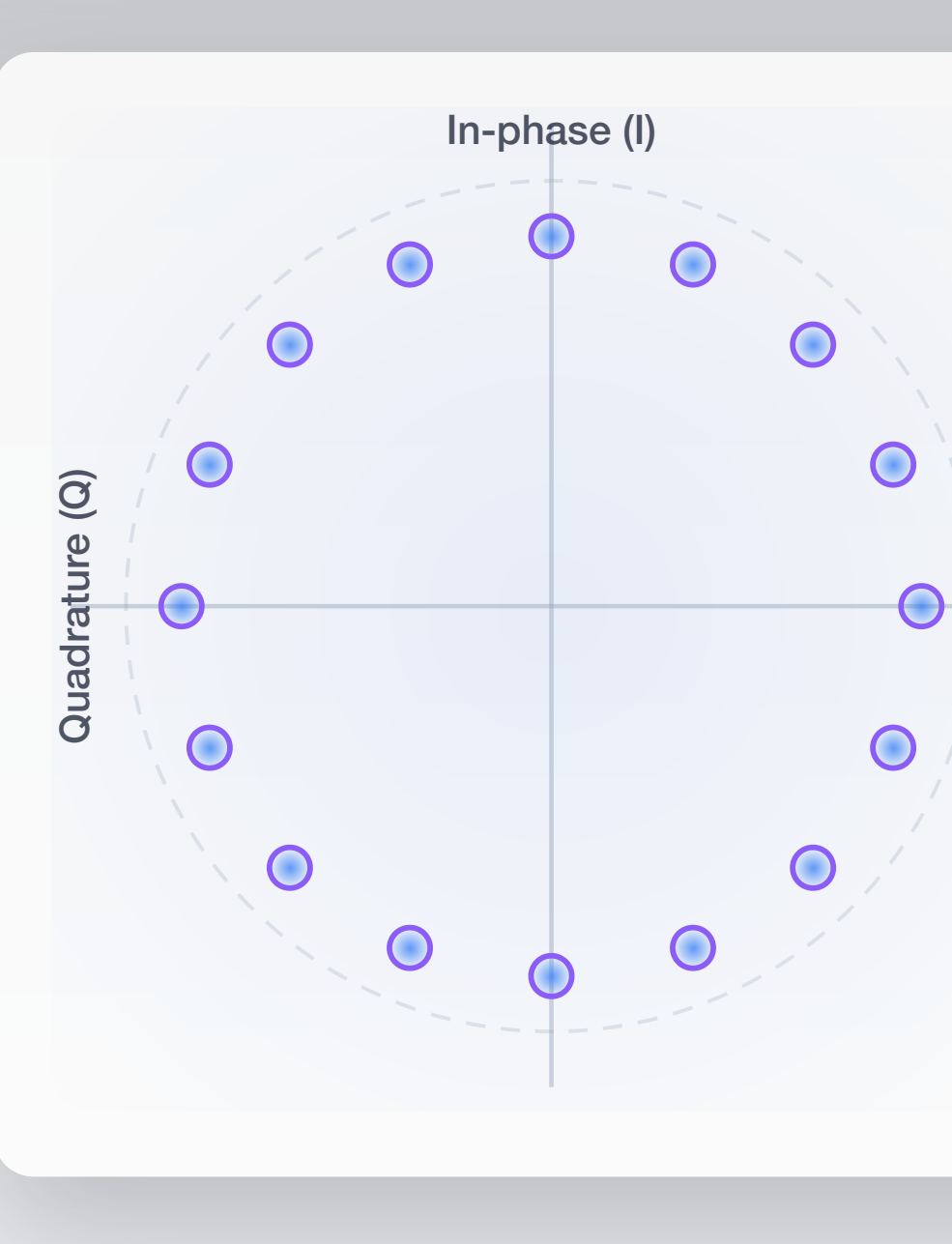
**Huffman Sweet Spot**
Natural-language skew: frequent vowels and digrams drive gains for Huffman, while limited phrase reuse leaves LZW with modest extra savings.
Use this text

**High-Entropy Noise**
A synthetic bit soup with symbols used uniformly at random—both compressors struggle because entropy is already near the 8-bit ceiling.
Use this text

## Introduction II

# How Digital Constellations Adapt to the Channel

Drag the points through phase-only, amplitude-and-phase, and ring-based layouts. Each constellation uses the same 16 symbols, but the geometry shifts to balance noise tolerance, non-linear hardware, and spectral efficiency.



PSK · QAM · APSK

**16-PSK · CONSTANT ENVELOPE**

All 16 symbols sit on one circle. Only the phase changes, which keeps the power amplifier happy but leaves neighbouring points tightly packed in angle.

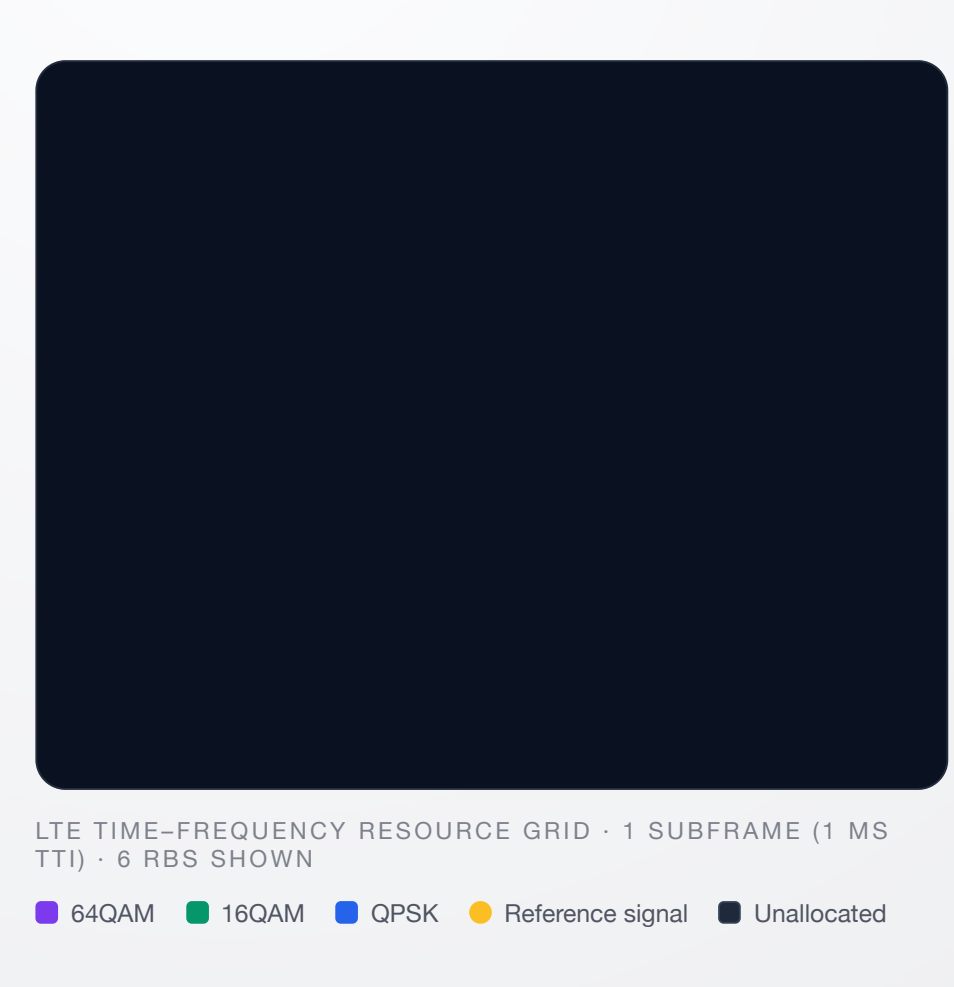
● Current symbol location ● Axes guide

Hover a point to see amplitude & phase

## 1.2 OFDM Modulation

# 1.2 LTE Requirement

LTE defines a set of performance targets that shaped every aspect of its air interface — from peak throughput and spectrum flexibility to mobility support and latency. The resource grid below shows one 1 ms subframe: 14 symbol intervals in time × 6 resource blocks (72 subcarriers) in frequency. Select a requirement to see how it drives allocation decisions.



LTE TIME–FREQUENCY RESOURCE GRID · 1 SUBFRAME (1 MS TTI) · 6 RBS SHOWN
64QAM
16QAM
QPSK
Reference signal
Unallocated

| PEAK THROUGHPUT | CHANNEL WIDTH | SPECTRAL EFFICIENCY |
|---|---|---|
| 326 Mbps | 1.4 – 20 MHz | ×3 – 4 DL |

| LATENCY | MOBILITY | COMPATIBILITY |
|---|---|---|
| < 5 ms | ≤ 350 km/h | All-IP + Legacy |

**Peak Data Rates**

Downlink reaches 326.4 Mbps with 4×4 MIMO and 64QAM; uplink peaks at 86.4 Mbps with a single transmit antenna.

Every available resource element carries 6 bits per symbol (64QAM). Four spatial MIMO layers then multiply that aggregate, delivering broadband-class throughput over a radio link. The grid shows a fully-loaded channel with 64QAM across all resource blocks.

| | |
|---|---|
| DOWNLINK (4×4 MIMO, 64QAM) | 326.4 MBPS |
| UPLINK (1×2 SIMO, 64QAM) | 86.4 MBPS |
| BITS PER SYMBOL (64QAM) | 6 BITS |

## 1.5 OFDM Explorer

1.5 OFDM EXPLORER

# From echoes to OFDM: why a radio channel becomes many slower lanes

A real wireless channel has memory. Reflections arrive late, smear one symbol into the next, and make a fast single-carrier stream painful to recover. This explorer builds the cure: longer OFDM symbols and a cyclic prefix that absorbs echoes before the receiver reads the useful part.

**1. Single carrier**
Short symbols suffer large relative ISI.

**2. OFDM**
Narrow subcarriers make symbols longer.

**3. Cyclic prefix**
The echo lands in a discarded buffer.

### The problem: echoes corrupt the next symbol

Imagine words shouted into a tunnel. The direct sound arrives first, but a delayed copy follows it. In a receiver that delayed copy is channel memory — part of the previous symbol arrives while the receiver is trying to read the next one.

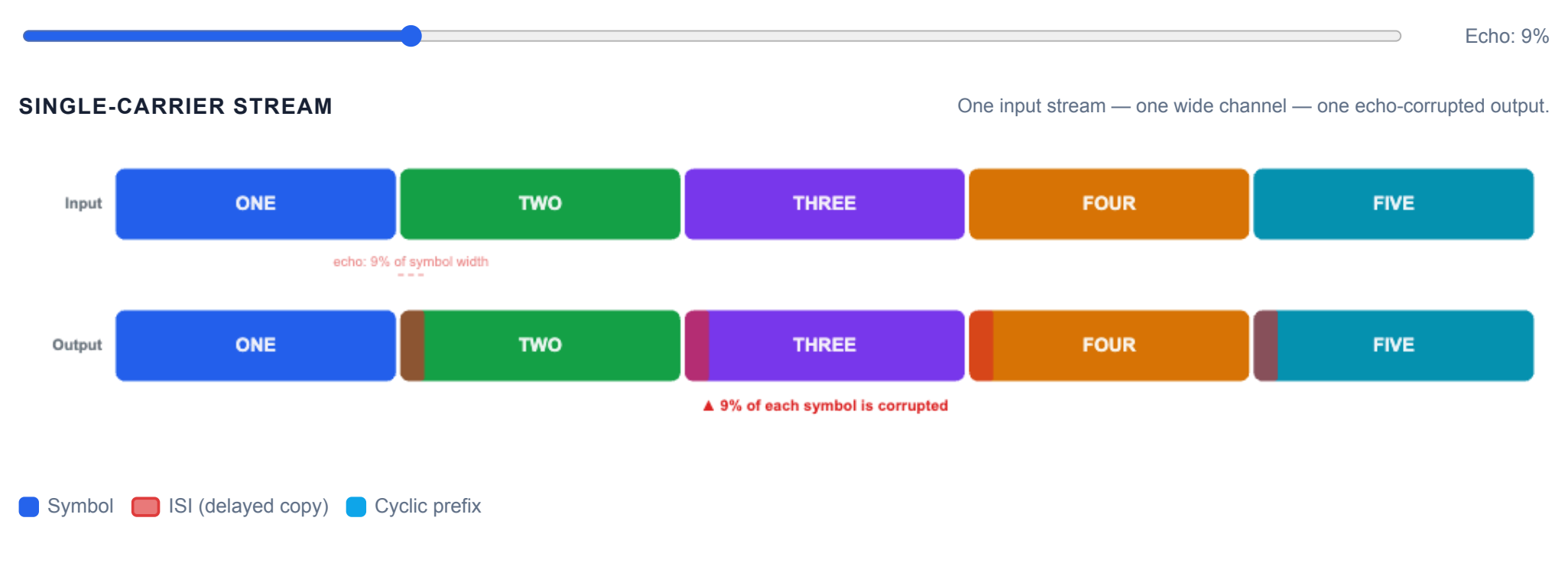
A fast single-carrier stream packs symbols tightly in time. The same echo therefore covers a large fraction of the next symbol, so the receiver sees a corrupted mixture rather than a clean word.

> Short symbols plus a fixed echo: the previous word visibly overlaps the next one.

Echo: 9%

SINGLE-CARRIER STREAM

One input stream — one wide channel — one echo-corrupted output.



### Technical lab: subcarriers, delay spread, and the prefix budget

The story above uses words as symbols. The actual transmitter uses complex modulation values. This lab keeps the engineering view: data subcarriers, pilot tones, guard bins, and a cyclic prefix. Move the sliders and watch whether the channel memory fits inside the prefix.

**Cyclic prefix length (samples)**

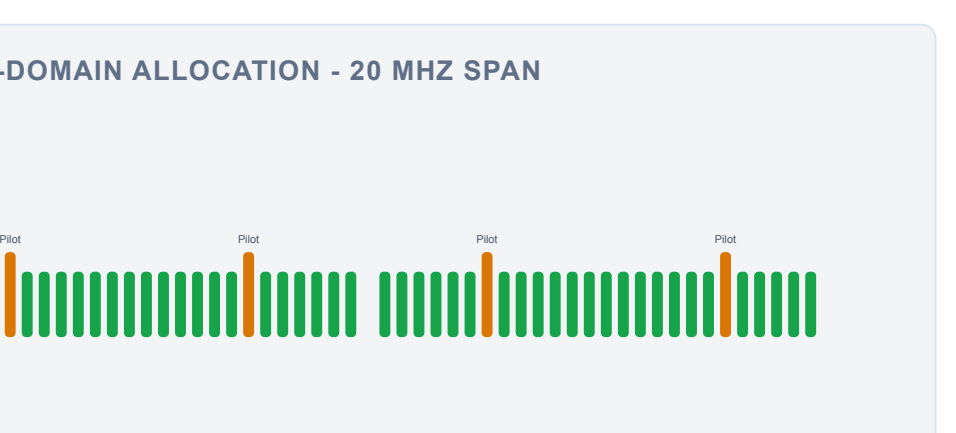
Prefix duration: 0.80 us

**Channel delay spread (0-1.2 us)**

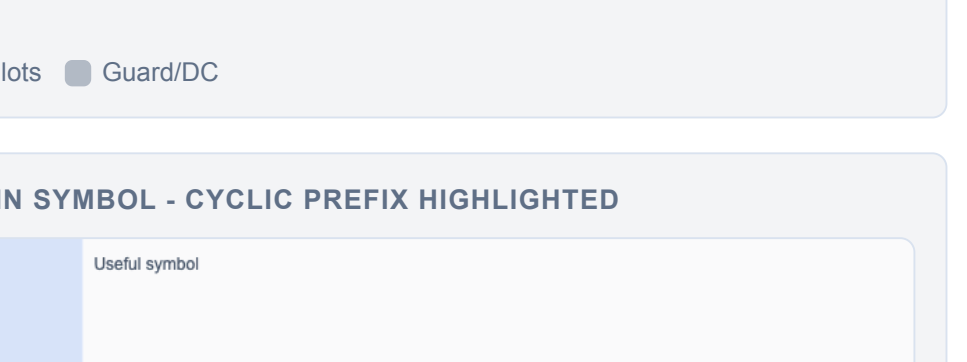
Delay spread: 0.60 us

**CP >= delay spread: echoes fit inside the discarded prefix.**

FREQUENCY-DOMAIN ALLOCATION - 20 MHZ SPAN



TIME-DOMAIN SYMBOL - CYCLIC PREFIX HIGHLIGHTED



**Useful symbol:** 3.20 us (64 samples)
**Cyclic prefix:** 0.80 us (16 samples)
**Leakage:** no residual ISI — echoes stay inside the prefix

## 2.2 LTE Bandwidth vs RBs

# How LTE Scales Bandwidth Without Touching Subcarrier Spacing

LTE locks the subcarrier spacing at **15 kHz**. Instead of squeezing carriers closer together, the network adds more Resource Blocks (RBs) as channel bandwidth grows. The interactive grid shows how many RBs fit inside each standardized channel size.

### 2.2 LTE STANDARDS

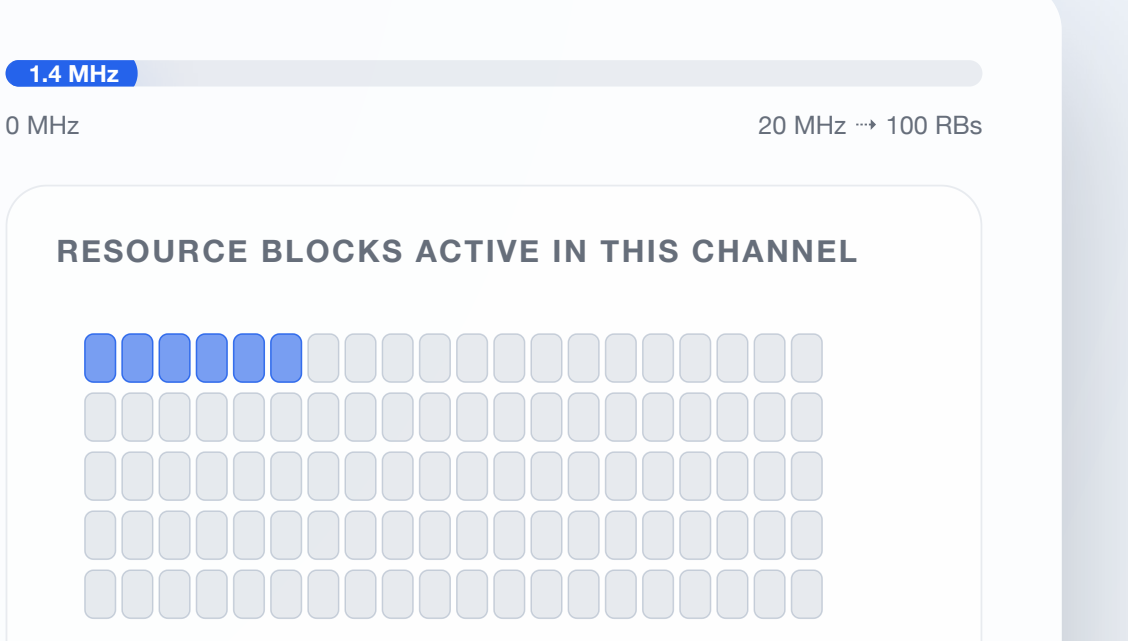
Every LTE Resource Block spans **12 subcarriers × 15 kHz**, and it is the smallest unit the scheduler can hand out in the uplink or downlink. A wider channel simply means more RBs on the dial—from just six in a narrow 1.4 MHz carrier up to one hundred in the full 20 MHz allocation.

**SUBCARRIER SPACING**
Fixed at 15 kHz no matter how wide the channel becomes. RB width ≈ 180 kHz.

1.4 MHz · 3 MHz · 5 MHz · 10 MHz · 15 MHz · 20 MHz

Channel: 1.4 MHz — Active RBs: 6

Play timeline — Used for guard-band friendly deployments and M2M/IoT.



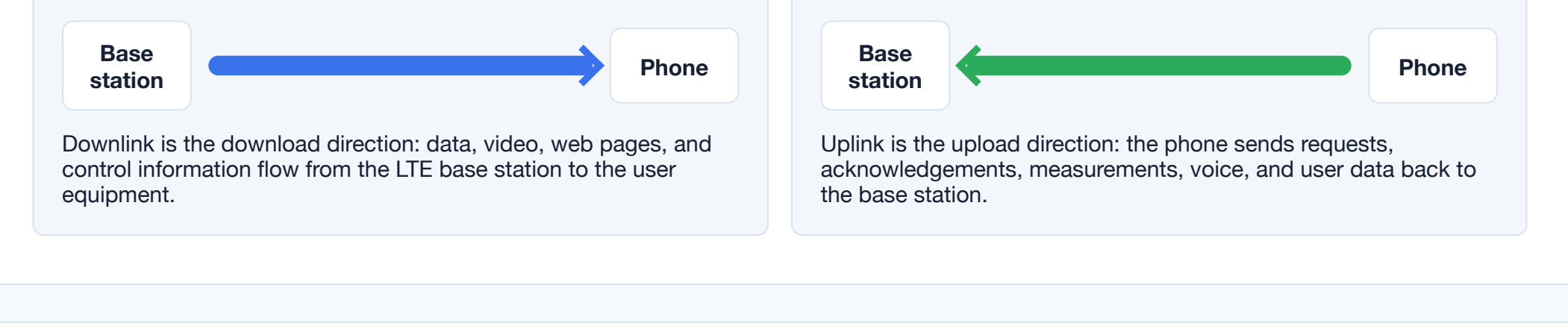
| | |
|---|---|
| Channel bandwidth | RBs available |
| 1.4 MHz | 6 |
| Subcarriers total | Total subcarrier span |
| 72 | 1.08 MHz ≈ user data |

## 2.3 LTE Duplexing

2.3 LTE DUPLEXING

# Uplink, Downlink, and how LTE keeps both directions from colliding

After OFDM and Resource Blocks, LTE still needs a basic traffic rule: the base station sends data to the phone, and the phone sends data back. Duplexing decides how those two directions share radio resources.

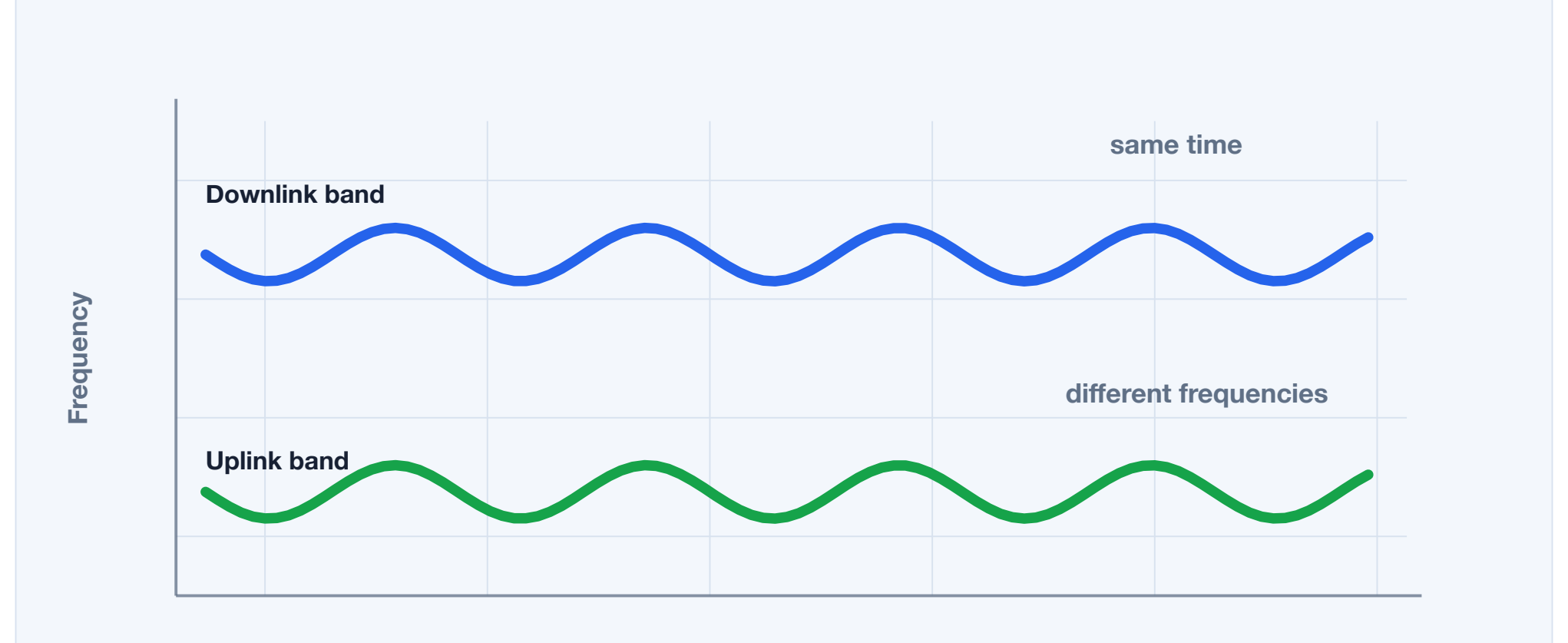


Downlink is the download direction: data, video, web pages, and control information flow from the LTE base station to the user equipment.

Uplink is the upload direction: the phone sends requests, acknowledgements, measurements, voice, and user data back to the base station.

**FDD**
Frequency Division Duplex: same time, different frequency bands.

**TDD**
Time Division Duplex: same frequency band, different time slots.



> **FDD separates directions by frequency: downlink and uplink can run at the same time because they occupy different bands.**

**ADVANTAGES**
- Downlink and uplink can be active simultaneously.
- Works naturally with paired spectrum allocations.
- Predictable timing because each direction has its own band.

**LIMITATIONS**
- Requires two separate frequency bands.
- Less flexible when traffic is asymmetric, such as heavy download and light upload.
- Needs strong separation between transmit and receive bands.

Duplexing answers which direction can use the radio resource. Once a resource is part of the downlink, LTE can decide which phone gets which Resource Blocks. That is the setup for OFDMA in section 2.6.

## 2.6 OFDMA Downlink Scheduler

2.6 OFDM/OFDMA DOWNLINK

# OFDMA is how the base station shares the downlink grid between phones

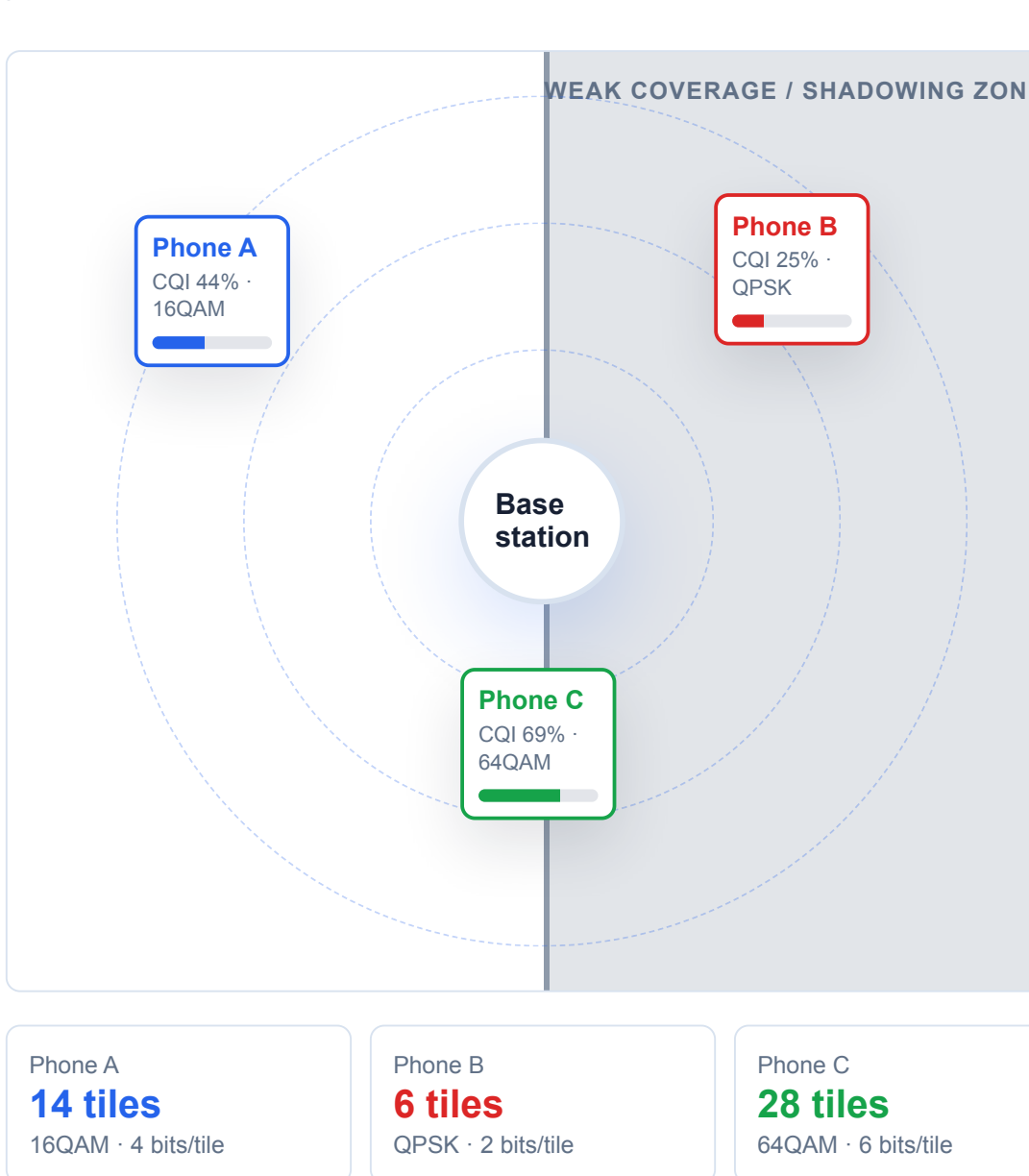
OFDM gives us a grid of subcarrier-like tiles. OFDMA turns that grid into a shared downlink: each phone hears the same wide signal, but the scheduler assigns different data tiles to different phones according to channel quality.

### Simplified downlink grid

This demo uses a simplified 64-tile OFDM grid. Only 48 tiles carry user data; the remaining tiles represent pilot/reference, guard, and control overhead.

Before a phone decodes the data grid, it receives a scheduling grant: a short control instruction that says which assigned data tiles to read and which modulation to expect.

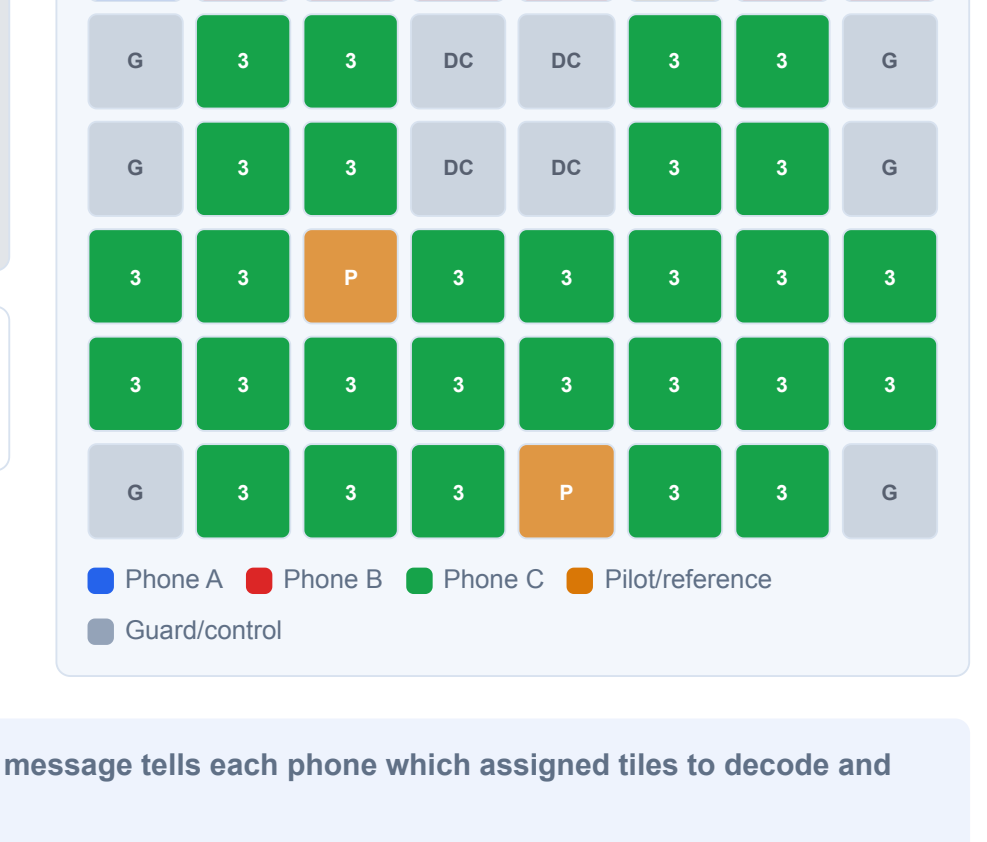
Drag the phones. Distance from the base station lowers channel quality. The gray half adds a shadowing penalty, so two phones at the same radius can still receive different allocations.



| Phone A | Phone B | Phone C |
|---|---|---|
| 14 tiles | 6 tiles | 28 tiles |
| 16QAM · 4 bits/tile | QPSK · 2 bits/tile | 64QAM · 6 bits/tile |

DOWNLINK CONTROL / SCHEDULING GRANTS

**Phone A**: decode data tiles 1-14 using 16QAM (4 bits/tile).
**Phone B**: decode data tiles 15-20 using QPSK (2 bits/tile).
**Phone C**: decode data tiles 21-48 using 64QAM (6 bits/tile).



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| G | 1 | 1 | P | 1 | 1 | 1 | G |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 2 | 2 | 2 | P | 2 | 2 |
| G | 3 | 3 | DC | DC | 3 | 3 | G |
| G | 3 | 3 | DC | DC | 3 | 3 | G |
| 3 | 3 | P | 3 | 3 | 3 | 3 | 3 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| G | 3 | 3 | 3 | P | 3 | 3 | G |

🟦 Phone A 🟥 Phone B 🟩 Phone C 🟧 Pilot/reference ⬜ Guard/control

> **OFDM builds the tiles. OFDMA assigns the data tiles. The downlink control message tells each phone which assigned tiles to decode and which modulation setting to use.**

## 2.7 Different Modulation Schemes

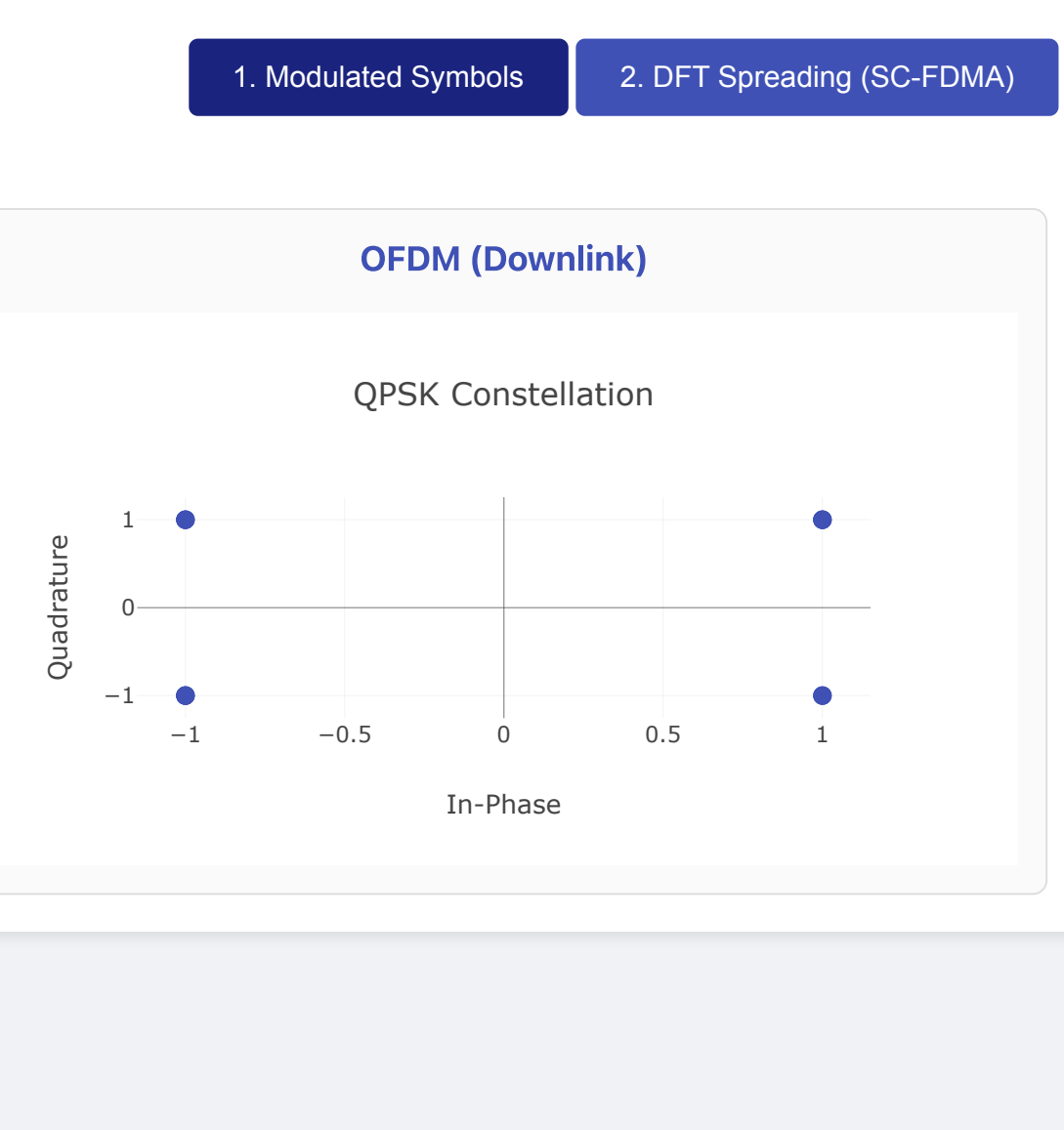
# OFDM vs. SC-FDMA

In LTE, the downlink (from base station to user) and uplink (from user to base station) use different technologies. The downlink relies on **OFDM (Orthogonal Frequency-Division Multiplexing)**, while the uplink adopts **SC-FDMA (Single-Carrier Frequency Division Multiple Access)**. The primary reason is that OFDM signals have a high **Peak-to-Average Power Ratio (PAPR)**, which requires more expensive and less power-efficient amplifiers on mobile devices. SC-FDMA was introduced to solve this issue, offering lower PAPR while keeping many of OFDM's benefits.

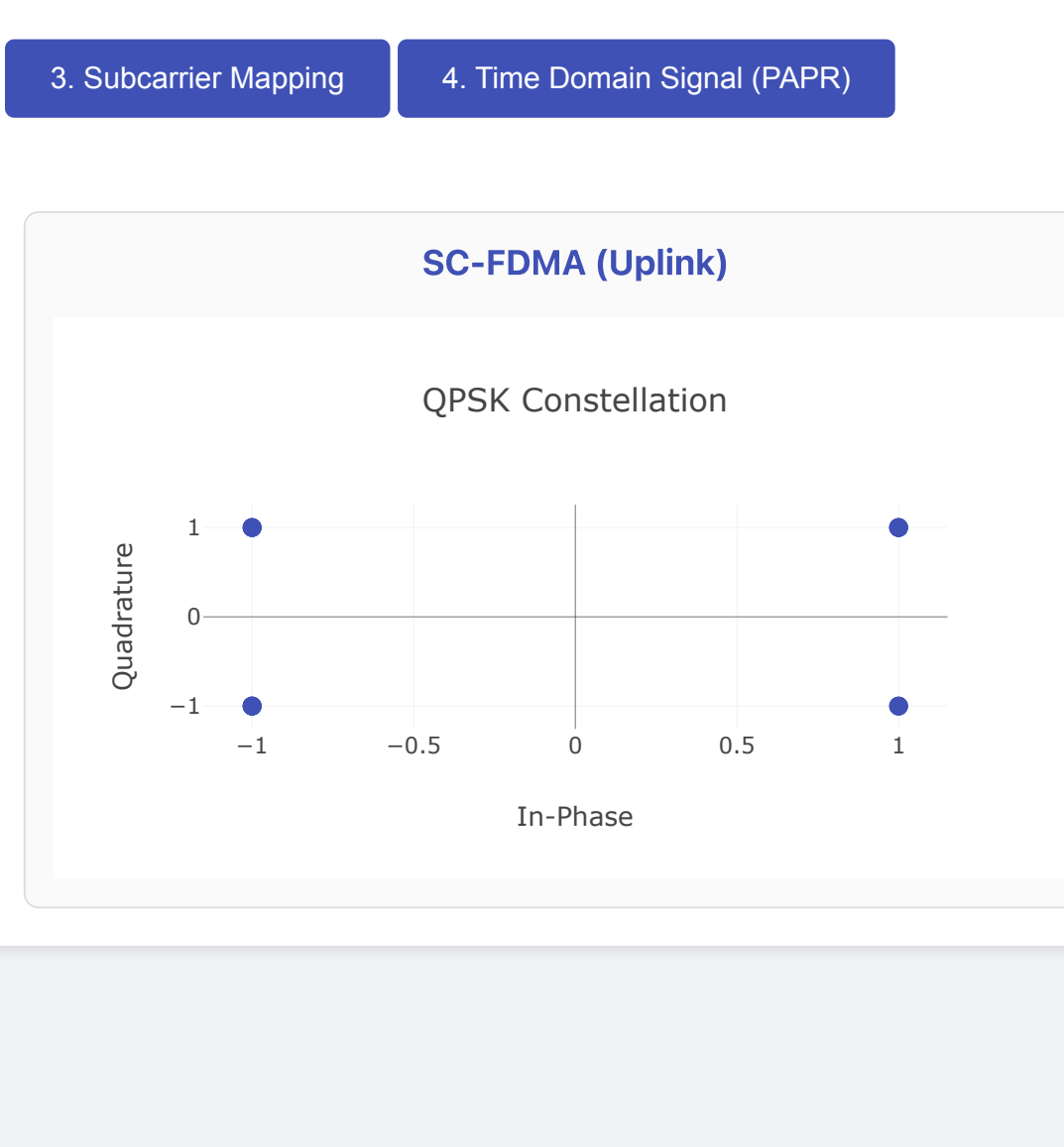
This visualization interactively demonstrates the signal generation process for both and highlights the key differences.

1. Modulated Symbols · 2. DFT Spreading (SC-FDMA) · 3. Subcarrier Mapping · 4. Time Domain Signal (PAPR)

**OFDM (Downlink)**



**SC-FDMA (Uplink)**



## 3.2 Transmission Reception Layer

# Transmission & Reception Layer: Quick Explorer

A tactile view of how LTE juggles physical signals, channels, and duplexing. Tap through the modules to see how synchronization beacons, control traffic, and user payloads share the air, whether spectrum is split (FDD) or traded in time (TDD).

### SIGNALS VS CHANNELS

Physical Signals · Physical Channels

Physical-layer signals are the beacons the UE listens for first; once synchronized, mapped channels keep control loops and payload streaming.

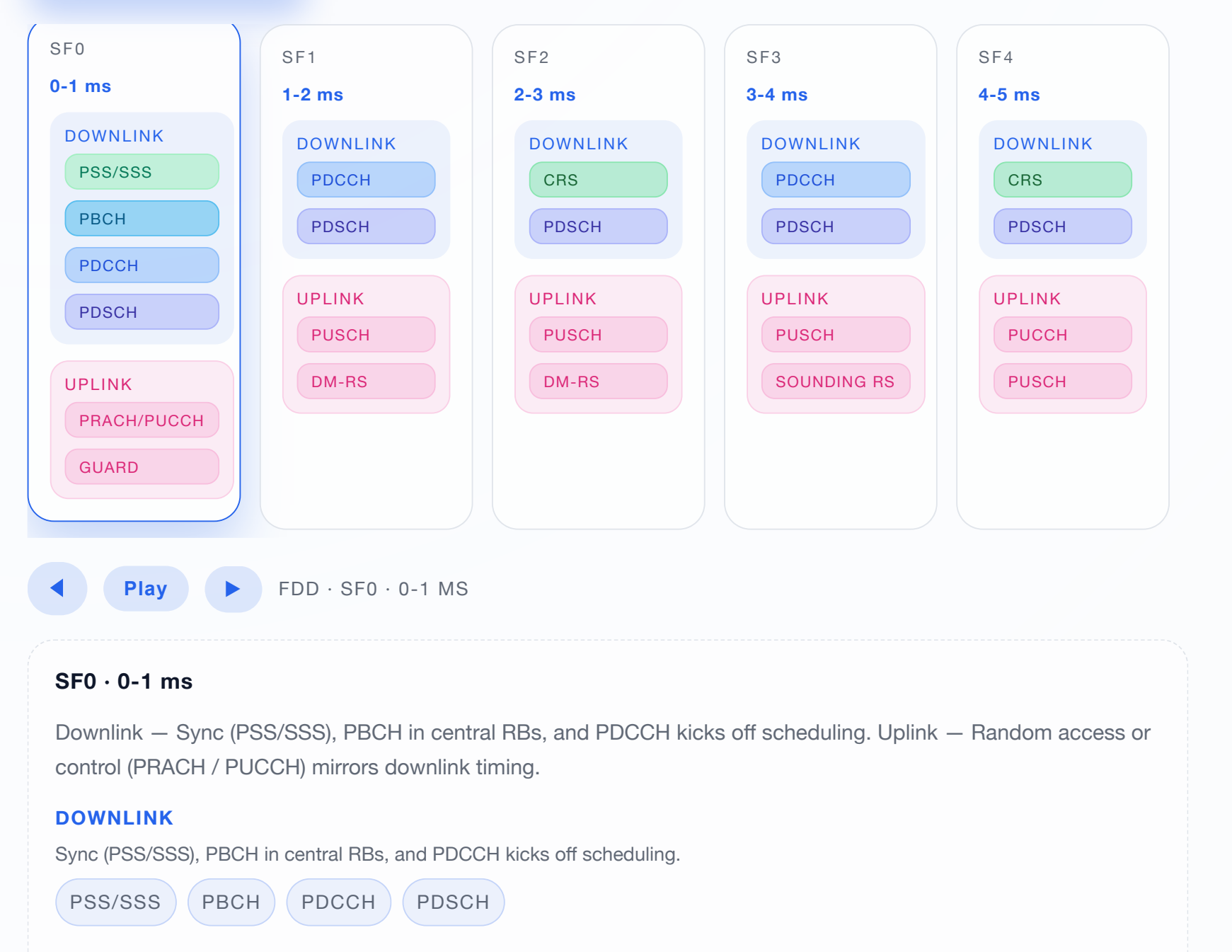
**PSS / SSS**
Twin sync bursts in subframes 0 and 5 let the UE lock timing and grab the physical cell ID.

**Reference Signals**
Dense pilot tones woven into every slot so receivers can gauge fading and fix amplitude/phase drift.

**Uplink Demod/Sounding RS**
UE transmissions aligned to the eNB; sounding probes keep uplink channel estimates fresh.

### FRAME FLOW

FDD — Split Spectrum · TDD — Split Time



◀ Play ▶ FDD · SF0 · 0-1 MS

**SF0 · 0-1 ms**

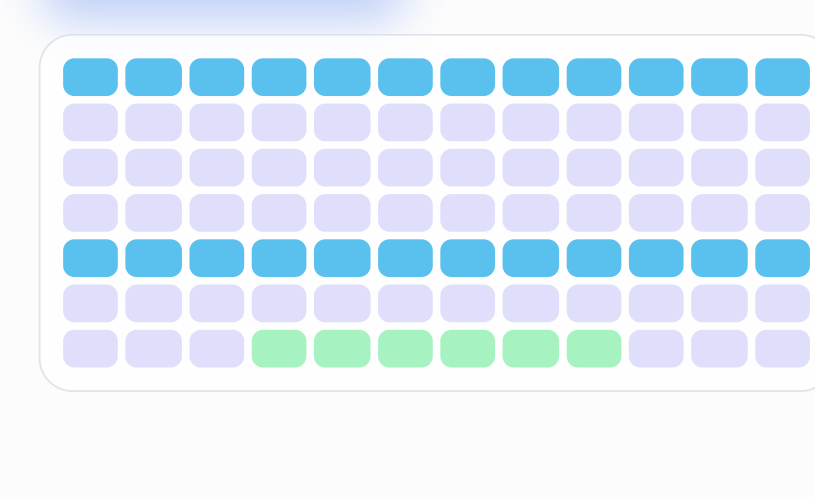
Downlink — Sync (PSS/SSS), PBCH in central RBs, and PDCCH kicks off scheduling. Uplink — Random access or control (PRACH / PUCCH) mirrors downlink timing.

**DOWNLINK**
Sync (PSS/SSS), PBCH in central RBs, and PDCCH kicks off scheduling.
PSS/SSS · PBCH · PDCCH · PDSCH

**UPLINK**
Random access or control (PRACH / PUCCH) mirrors downlink timing.
PRACH/PUCCH · GUARD

### RESOURCE BLOCK ANATOMY

Normal Cyclic Prefix · Extended Cyclic Prefix



- Cell-specific reference symbols anchor channel estimation.
- PSS/SSS bursts appear in central subcarriers of subframes 0 and 5.
- PDSCH / PUSCH resource elements ferry user data once control is set.

Normal CP → 12 subcarriers × 7 symbols = 84 resource elements.